

Full-Scale Hybrid Attention Network for Small ROI Medical Image Segmentation

Detecting thin retinal vessels using attention-guided multi-scale feature fusion — a deep learning approach for clinical eye disease screening.

ABSTRACT

Accurate segmentation of retinal blood vessels is essential for early detection of diseases such as diabetic retinopathy and glaucoma. This remains challenging because vessels are extremely thin, vary greatly in scale, and have low contrast against the surrounding background. Standard segmentation models frequently fail to capture these fine structures.

This project proposes **FSHA-Net (Full-Scale Hybrid Attention Network)**, a deep learning model that combines full-scale skip connections (U-Net 3+ style) with soft attention gates and deep supervision to improve segmentation of small, clinically significant vessel regions. A baseline **U-Net** is trained under identical conditions for fair architectural comparison. Both models are evaluated on the **DRIVE dataset** using Dice Score, IoU, HD95, Sensitivity, and Specificity. Results will demonstrate that attention-guided multi-scale feature fusion meaningfully improves detection of fine vessel structures over the U-Net baseline.

INTRODUCTION

Eye diseases such as diabetic retinopathy affect hundreds of millions of people globally, and a large fraction go undiagnosed until irreversible damage has occurred. Key diagnostic indicators include the width, branching pattern, and density of retinal blood vessels — all of which can be assessed from fundus photographs.

Manual examination of retinal images is slow, subjective, and requires trained specialists. Automating this with deep learning can make screening faster and more accessible, particularly in regions where specialist eye care is scarce.

The core difficulty is that retinal vessels — especially thin capillaries — occupy a very small fraction of the image. Standard U-Net models miss these structures because skip connections carry information from only one matching encoder scale, limiting the model's ability to reason across multiple levels of detail.

FSHA-Net addresses this by routing features from all encoder levels into every decoder node, combined with soft attention gates that suppress background regions and focus on actual vessel pixels. Deep supervision at multiple decoder scales provides additional gradient signal during training, improving convergence on this small dataset (only 20 training images).

BRIEF LITERATURE SURVEY

U-Net (Ronneberger et al., 2015) — Encoder-decoder with skip connections. Became the standard for medical image segmentation. Limitation: each decoder node receives features from only one matching encoder scale.

Attention U-Net (Oktay et al., 2018) — Adds soft attention gates to U-Net skip connections. Gates suppress irrelevant background regions. Shows consistent improvement on small ROI segmentation tasks.

U-Net 3+ (Huang et al., 2020) — Redesigns skip connections so every decoder node aggregates features from all encoder scales and all previous decoder nodes. Full-scale feature fusion significantly improves fine-structure detection.

Deep Supervision (Lee et al., 2015) — Attaches output heads at intermediate decoder nodes to provide direct gradient signal at multiple scales. Particularly effective when training data is limited.

FSHA-Net (this work) — Combines U-Net 3+ full-scale skips with attention gates and deep supervision. The hybrid design targets the specific challenge of small, low-contrast vessel structures in retinal images.

DATASET DESCRIPTION

DRIVE — Digital Retinal Images for Vessel Extraction

The **DRIVE dataset (Digital Retinal Images for Vessel Extraction)** is a standard benchmark for retinal vessel segmentation, collected as part of a diabetic retinopathy screening programme in the Netherlands. Around 400 diabetic patients aged 25–90 participated; 40 retinal images were selected for research. Most images show no signs of retinopathy and a few contain mild abnormalities — reflecting a realistic clinical distribution rather than a curated disease-only set.

Property	Value
Total images	40 colour fundus images
Training set	20 images
Test set	20 images
Image resolution	565×584 px (cropped from 768×584)
Camera	Canon CR5 non-mydratic 3CCD
Field of view	45° FOV diameter ≈ 540 px
Format	JPEG compressed
Annotations	Hand-labelled by trained observers
Second observer	Available as upper-bound reference
Patient age range	25 – 90 years (diabetic screening)

Why DRIVE is challenging for segmentation:

- Vessel widths range from 1–2 px (thin capillaries) to several pixels (arteries)
- Low contrast between thin vessels and surrounding background
- Bright structures (optic disc) introduce false positive regions
- Only 20 training images — very limited data for deep learning
- Class imbalance: vessel pixels are a tiny fraction of the total image area

METHODOLOGY

Preprocessing Pipeline

- Resize all images to 256×256 px
- Normalise with ImageNet mean/std: $\mu=(0.485,0.456,0.406)$, $\sigma=(0.229,0.224,0.225)$
- Binarise masks: any non-zero pixel $\rightarrow$ 1 (vessel), else 0 (background)

Augmentation (Albumentations, applied identically to image and mask)

- HorizontalFlip ($p=0.5$)
- Rotate $\pm 30^\circ$ ($p=0.5$)
- RandomCrop $\rightarrow$ resize back to 256×256 ($p=0.5$)
- RandomBrightnessContrast ($p=0.4$)

Baseline Model: U-Net

Standard encoder-decoder with skip connections. Encoder: 5 levels doubling channels (64 $\rightarrow$ 128 $\rightarrow$ 256 $\rightarrow$ 512 $\rightarrow$ 1024). Decoder: bilinear upsample + concat skip + DoubleConv at each level. Output: 1×1 Conv $\rightarrow$ raw logits.

Proposed Model: FSHA-Net

Three-component design: (1) same 5-level encoder as U-Net; (2) U-Net 3+ style full-scale skip connections — every decoder node receives features from all 4 encoder levels; (3) soft attention gate on every skip connection before concatenation; (4) deep supervision with three output heads.

FSHA-NET ARCHITECTURE

Layer	Output Size	Channels	Operation	
Input	256×256	3	RGB image	
Encoder E1	256×256	64	DoubleConv	
Encoder E2	128×128	128	MaxPool + DoubleConv	
Encoder E3	64×64	256	MaxPool + DoubleConv	
Encoder E4	32×32	512	MaxPool + DoubleConv	
Bottleneck	16×16	1024	MaxPool + DoubleConv	
Decoder D4	32×32	512	FSNode + AttGate×4	
Decoder D3	64×64	256	FSNode + AttGate×4	
Decoder D2	128×128	128	FSNode + AttGate×4	
Decoder D1	256×256	64	FSNode + AttGate×4	
Out (main)	256×256	1	Conv1×1 + Sigmoid	
Out (aux1)	256×256	1	Conv1×1 + Upsample	
Out (aux2)	256×256	1	Conv1×1 + Upsample	

Attention Gate: For each skip feature x and gating signal g , $\alpha = \sigma(W_x(x) + W_g(g))$. The attended output is $x \cdot \alpha$, which suppresses background regions and highlights vessel-relevant features before concatenation in the decoder.

Full-Scale Skip: Instead of connecting encoder level i only to decoder level i (U-Net), every decoder node receives all 4 encoder feature maps, resized to the target resolution and projected to a uniform channel width before fusion. This gives each decoder node access to both fine-grained and semantic-level information.

LOSS FUNCTIONS

Loss Component	Formula	Weight	
Dice Loss	$1 - 2 P \cap G / (P + G)$	—	
Focal Loss	$-\alpha(1-p_{\text{vessel}})^{\gamma} \cdot \log(p_{\text{vessel}})$	0.5	
Seg Loss	Dice + 0.5×Focal	—	
L_main	SegLoss(main output)	0.5	
L_aux1	SegLoss(d2 output ↑256)	0.3	
L_aux2	SegLoss(d3 output ↑256)	0.2	
L_total	$0.5 \cdot L_{\text{main}} + 0.3 \cdot L_{\text{aux1}} + 0.2 \cdot L_{\text{aux2}}$	1.0	

Dice Loss directly optimises the Dice coefficient, making it naturally suited to class-imbalanced segmentation. **Focal Loss** downweights easy background pixels and forces the model to focus on hard, thin vessel regions. Their combination is effective for small ROI segmentation.

TRAINING CONFIGURATION

Hyperparameter	Value	Reason
Optimiser	Adam	Adaptive LR, works well for seg.
Learning rate	1e-4	Standard starting point
Weight decay	1e-5	Mild L2 regularisation
Batch size	4	Fits typical GPU memory
Input size	256×256	Balance detail vs. memory
LR schedule	ReduceOnPlateau (×0.5)	Adapts to plateaus
Grad clipping	1.0	Prevents exploding gradients
Early stopping	patience=15	Avoids overfitting on 20 samples
Epochs	50 max	Sufficient for DRIVE
Augmentation	Flip, Rotate, Crop, BCJitter	Prevents overfitting

LR Scheduler: ReduceLROnPlateau halves the learning rate after 7 consecutive epochs with no improvement in validation Dice. This prevents oscillation around local minima late in training.

Early Stopping: Training halts after 15 epochs of no val-Dice improvement. Critical here because DRIVE has only 20 training samples — overfitting risk is high without this safeguard.

Gradient Clipping: Max norm = 1.0. Prevents exploding gradients that can occur in deep networks on very small datasets.

EVALUATION METRICS

Metric	Formula	Range	Better
Dice Score	$2 P \cap G / (P + G)$	0 – 1	Higher
IoU (Jaccard)	$ P \cap G / P \cup G $	0 – 1	Higher
HD95	95th pct surface dist.	px	Lower
Sensitivity	$TP / (TP + FN)$	0 – 1	Higher
Specificity	$TN / (TN + FP)$	0 – 1	Higher

Why HD95 and not HD100? The 95th-percentile Hausdorff Distance is more robust than the maximum because it is not affected by single outlier pixels. It measures how well the predicted boundary matches the true vessel boundary at the 95th-worst point.

Why Sensitivity matters more than Specificity in clinical screening: Missing a vessel (low sensitivity) means the model failed to detect a clinically relevant structure. A false positive (low specificity) is less dangerous because it prompts further review rather than missing a diagnosis.

EXPECTED RESULTS & CRITICAL ANALYSIS

Based on published work on DRIVE, well-trained models typically achieve Dice scores of 0.80–0.82 and IoU of 0.68–0.72 on the test set. Human inter-observer agreement sits around Dice $\approx$ 0.80.

FSHA-Net vs U-Net (expected):

- Higher Dice and IoU on thin vessel structures due to full-scale fusion
- Lower HD95 due to attention gates producing cleaner vessel boundaries
- Better sensitivity (fewer missed capillaries)
- Slightly slower inference due to additional attention computation

Potential failure modes and analysis:

- **Low contrast regions:** Both models may miss vessels near the image periphery where contrast drops significantly.
- **Optic disc interference:** The bright optic disc region can cause false positives. Attention gates partially mitigate this.
- **Class imbalance:** With only ~10% of pixels being vessel, the Focal Loss component is critical to prevent the model from predicting all-background.
- **Small dataset:** Only 20 training images. Augmentation and early stopping are essential to prevent overfitting.

PROJECT STRUCTURE & USAGE

File layout:

```
data_loader.py – DRIVE dataset, stem-based matching, augmentation
models/unet.py – Full U-Net baseline
models/fsha_net.py – FSHA-Net with attention + deep supervision
loss/loss.py – DiceLoss, FocalLoss, SegLoss, DeepSupervisionLoss
metrics/metrics.py – Dice, IoU, HD95, Sensitivity, Specificity
utils.py – Device, checkpoints, visualisation, AverageMeter
train.py – Training loop (LR sched, grad clip, early stop)
eval.py – Evaluation + 5-metric comparison table
```

Run commands:

```
pip install torch torchvision albumentations numpy Pillow scipy
python train.py --model unet --epochs 50 --batch_size 4
python train.py --model fshanet --epochs 50 --batch_size 4
python eval.py --model both
```

CONCLUSION & FUTURE SCOPE

This project implements and compares two deep learning architectures for retinal vessel segmentation on the DRIVE dataset. The proposed FSHA-Net combines full-scale skip connections, soft attention gates, and deep supervision to address the core challenge of detecting thin, low-contrast vessel structures.

Future directions:

- Pre-train the encoder on ImageNet for better feature initialisation
- Apply test-time augmentation (TTA) to improve prediction stability
- Extend to 3D volumetric data (e.g. brain vessel MRI)
- Explore transformer-based encoders (e.g. Swin-UNet)
- Deploy as a clinical screening tool with a confidence threshold

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